

Haodong Yan

☎ (+86) 15829011780 | ✉ hyan758@connect.hkust-gz.edu.cn

Homepage | Google Scholar

Education

The Hong Kong University of Science and Technology (Guangzhou)

Ph.D. in Intelligent Transportation

Guangzhou, China

Sep. 2024 – Present

Xi'an Jiaotong University

Master of Mechanical Engineering

Rank: 1/281

GPA: 3.73/4.0, Average Score: 89.68

Xi'an, China

Sep. 2021 – Jul. 2024

Xi'an Jiaotong University

Bachelor of Mechanical Engineering

Xi'an, China

Sep. 2017 – Jan. 2021

GPA: 3.75/4.3, Average Score: 88.75

Xi'an Jiaotong University

Second Bachelor of Computer Science and Technology

Xi'an, China

Jul. 2019 – Jul. 2021

Selected Publications

- [1] **Haodong Yan**, Zhide Zhong, Jiaguan Zhu, Junjie He, Weilin Yuan, Wenxuan Song, Xin Gong, Yingjie Cai, Guanyi Zhao, Xu Yan, Bingbing Liu, Ying-Cong Chen, Haoang Li, “S-VAM: Shortcut Video-Action Model by Self-Distilling Geometric and Semantic Foresight,” *ECCV*, 2026.
- [2] **Haodong Yan***, Hang Yu*, Zhide Zhong*, Weilin Yuan, Xin Gong, Zehang Luo, Chengxi Heyu, Junfeng Li, Wenxuan Song, Shunbo Zhou, Haoang Li, “Open-world Hand-Object Interaction Video Generation Based on Structure and Contact-aware Representation,” *CVPR*, 2026.
- [3] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Xiangchen Liu, Xin Gong, Tianran Zhang, Wenxuan Song, Jiayi Chen, Xinhua Zheng, Hesheng Wang, Haoang Li, “FlowVLA: Visual Chain of Thought-based Motion Reasoning for Vision-Language-Action Models,” Preprint, under review at *IEEE Transactions on Robotics*.
- [4] **Haodong Yan**, Pengxu Hou, Zhide Zhong, Xinhua Zheng, Zhe Liu, Hesheng Wang, Haoang Li, “CARE: Contextually-Aligned and Realistic 4D Scene Generation from a Single Image and Text,” Preprint, under review at *IEEE Transactions on Image Processing*.
- [5] **Haodong Yan**, Jiaguan Zhu, Mingyuan Jia, Ruiqing Yin, Junjie He, Zhide Zhong, Junfeng Li, Jinxuan Lu, Hengtao Li, Tianran Zhang, Jiayi Chen, Wenxuan Song, Wen Chen, Yuxiang Gao, Haoang Li, “Is Forward Prediction Enough? Physical State Grounding for JEPA World Models,” *arXiv*, 2026.
- [6] **Haodong Yan**, Junfeng Li, Junjie He, Zhide Zhong, MingMing Yu, Wenxuan Song, Jiaguan Zhu, Yangyang Zheng, Yuqiao Du, Jiadi You, Yingjie Cai, Xu Yan, Guanyi Zhao, Bingbing Liu, Haoang Li, “Robust-WAM: Bridging Generative Pretraining and Semantic Foresight in World-Action Models,” *arXiv*, 2026.
- [7] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Junjie He, Tianran Zhang, Haoang Li, “VLA-OPD: Bridging Offline SFT and Online RL for Vision-Language-Action Models via On-Policy Distillation,” *arXiv*, 2026.
- [8] Zhide Zhong, Junfeng Li, Junjie He, **Haodong Yan**, Xin Gong, Guanyi Zhao, Yingjie Cai, Jiantao Gao, Xu Yan, Bingbing Liu, Yingcong Chen, Liuqing Yang, Haoang Li, “DualCoT-VLA: Visual-Linguistic Chain of Thought via Parallel Reasoning for Vision-Language-Action Models,” *arXiv*, 2026.
- [9] Huashuo Lei, Wenxuan Song, Huarui Zhang, Jieyuan Pei, Jiayi Chen, **Haodong Yan**, Han Zhao, Pengxiang Ding, Zhipeng Zhang, Lida Huang, Donglin Wang, Yan Wang, Haoang Li, “RoboMemArena: A Comprehensive and Challenging Robotic Memory Benchmark,” *arXiv*, 2026.

- [10] Junfeng Li, Junjie He, Zhide Zhong, Yangyang Zheng, Pingyue Sheng, Jiayu Dong, Ruixin Li, **Haodong Yan**, Jiaguan Zhu, Tianran Zhang, Runze Yu, Wen Chen, Liuqing Yang, Yuxiang Gao, Haoang Li, “DyPES-VLA: Learning Shared Dynamics Priors and Embodiment-Specific Control for Cross-Embodiment Manipulation,” *arXiv*, 2026.
- [11] GigaWorld Team (incl. **Haodong Yan**), “GigaWorld-Policy-0.5: A Faster and Stronger WAM Empowered by AutoResearch,” *arXiv*, 2026.
- [12] Wenxuan Song, Jiayi Chen, Xiaoquan Sun, Huashuo Lei, Yikai Qin, Wei Zhao, Pengxiang Ding, Han Zhao, Tongxin Wang, Pengxu Hou, Zhide Zhong, **Haodong Yan**, Donglin Wang, Jun Ma, Haoang Li, “Rethinking the Practicality of Vision-language-action Model: A Comprehensive Benchmark and An Improved Baseline,” *ICRA*, 2026.
- [13] Wenxuan Song, Ziyang Zhou, Han Zhao, Jiayi Chen, Pengxiang Ding, **Haodong Yan**, Yuxin Huang, Feilong Tang, Donglin Wang, Haoang Li, “ReconVLA: Reconstructive Vision-Language-Action Model as Effective Robot Perceiver,” *AAAI*, 2026 (Outstanding Paper Award).
- [14] **Haodong Yan***, Zhiming Hu*, Syn Schmitt, Andreas Bulling, “GazeMoDiff: Gaze-guided Diffusion Model for Stochastic Human Motion Prediction,” *Pacific Graphics*, 2024.
- [15] Cheng Lei, Junpeng Hu, **Haodong Yan**, Mariia Gladkova, Tianyu Huang, Yun-Hui Liu, Daniel Cremers, Haoang Li, “Physically-Based Photometric Bundle Adjustment in Non-Lambertian Environments,” *IROS*, 2024.
- [16] **Haodong Yan**, Fudong Li, Jinglong Chen, Zijun Liu, Jun Wang, Yong Feng, Xinwei Zhang, “A Graph Embedded in Graph Framework with Dual-sequence Input for Efficient Anomaly Detection of Complex Equipment under Insufficient Samples,” *Reliability Engineering & System Safety*, 2023.
- [17] **Haodong Yan**, Zijun Liu, Jinglong Chen, Yong Feng, Jun Wang, “Memory-augmented Skip-connected Autoencoder for Unsupervised Anomaly Detection of Rocket Engines with Multi-source Fusion,” *ISA Transactions*, 2023.
- [18] **Haodong Yan**, Jun Wang, Jinglong Chen, Zijun Liu, Yong Feng, “Virtual Sensor-based Imputed Graph Attention Network for Anomaly Detection of Equipment with Incomplete Data,” *Journal of Manufacturing Systems*, 2022.

* equal contribution.

Work Experience

GigaAI

Internship

Beijing, China

May 2026 – Present

- World-Action Model (WAM) foundation model training (GigaWorldPolicy)

Institute for Visualisation and Interactive Systems, University of Stuttgart

Internship

Stuttgart, Germany

Jul. 2023 – Oct. 2023

- Code implementation
- Paper writing

Computer Vision Group, Technical University of Munich

Remote Internship

Remote

Apr. 2023 – Dec. 2023

- Code implementation
- Paper writing
- Data processing

Honours and Awards

National Scholarship – Ministry of Education, China	2023
National Scholarship – Ministry of Education, China	2022
China Optics Valley Scholarship – Wuhan High-tech Zone Government	2022
Outstanding Graduates – Xi'an Jiaotong University	2021
HIWIN Elite Scholarship – HIWIN Technologies Corp.	2021
National First Prize – Chinese University Students Mechanical Innovation Competition	2020
National First Prize – Chinese Intelligent Robot Fighting Competition	2019

Skills

Programming: Python (PyTorch, OpenCV, Transformers), C++, L^AT_EX, Git.
Research Interests: Robot Learning (Video-Action Models, World Models), Video Generation Models.

Languages

English: IELTS Overall 7.0 (Listening 7.0, Reading 8.5, Writing 6.5, Speaking 6.0).
Mandarin Chinese: Native.